

CM0832 - MT5001 Elements of Set Theory

2. Relations, Functions and Orderings

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Outline

- Preliminaries
- Introduction
- Ordered Pairs
- Relations
- References

Preliminaries

Textbook

Karel Hrbacek and Thomas Jech ([1978] 1999). Introduction to Set Theory.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

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Introduction

Starting...

*“We begin our program of developing set theory as a **foundation for mathematics** by showing how various general mathematical concepts, such as relations, functions, and orderings **can be represented by sets.**” (p. 17)*

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Ordered Pairs

Definition (2)1.1

Let a and b be sets. We define the **ordered pair** of a and b , denoted (a, b) , where a is the **first coordinate** and b is the **second coordinate**, using Kuratowski (1921)'s definition, that is,

$$(a, b) \stackrel{\text{def}}{=} \{\{a\}, \{a, b\}\}.$$

Ordered Pairs

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Question

Is (a, b) a set?

Ordered Pairs

Example

We show that $(\emptyset, \{\emptyset\}) \neq (\{\emptyset\}, \emptyset)$.

$$(\emptyset, \{\emptyset\}) \stackrel{\text{def}}{=} \{ \{\emptyset\}, \{\emptyset, \{\emptyset\}\} \} \neq \{ \{\{\emptyset\}\}, \{\{\emptyset\}, \emptyset\} \} \stackrel{\text{def}}{=} (\{\emptyset\}, \emptyset).$$

Ordered Pairs

Example

Let a be a set. Then

$$\begin{aligned}(a, a) &\stackrel{\text{def}}{=} \{\{a\}, \{a, a\}\} \\ &\doteq \{\{a\}, \{a\}\} \\ &\doteq \{\{a\}\}.\end{aligned}$$

Ordered Pairs

Exercise [2.]1.1

Let a and b sets.

- (i) Prove that $(a, b) \in \mathcal{P}(\mathcal{P}(\{a, b\}))$.
- (ii) Prove that $a, b \in \bigcup(a, b)$.
- (iii) Prove that if $a \in A$ and $b \in A$, then $(a, b) \in \mathcal{P}(\mathcal{P}(A))$.

Ordered Pairs

Theorem [2.]1.8

Let a, a', b, b' be sets. Then,

$$(a, b) \doteq (a', b') \quad \text{if and only if} \quad a \doteq a' \text{ and } b \doteq b'.$$

Ordered Pairs

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Exercise

To give a different definition of ordered pair. Note that your definition must be a set and it must satisfy the Theorem [2.]1.8.

Ordered Pairs

Definitions

Let a , b , c and d sets. Then

$$(a, b, c) \stackrel{\text{def}}{=} ((a, b), c)$$

(ordered triples)

$$(a, b, c, d) \stackrel{\text{def}}{=} ((a, b, c), d)$$

(ordered quadruples)

$$(a) \stackrel{\text{def}}{=} a$$

(one-tuples)

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Relations

Definition [2.]2.1

A set R is a **binary relation** if and only if all elements of R are ordered pairs.

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Remark

*“A binary relation is, therefore, determined by specifying all ordered pairs of objects in that relation; **it does not matter by what property** the set of these ordered pairs is described.” (p. 19)*

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Notation

Let R be a binary relation. We write $(a, b) \in R$ or $a R b$.

Relations

Example (informal)

In the lecture.

Relations

Example (informal)

In the lecture.

Example (informal)

Let $\omega = \{0, 1, 2, \dots\}$. The identity relation on ω is defined by

$$\begin{aligned}\text{Id}_\omega &= \{ (n, n) \mid n \in \omega \} \\ &= \{ (0, 0), (1, 1), (2, 2), \dots \}.\end{aligned}$$

Relations

Example (definition without convention)

Let R be a binary relation. We define the following sets:

$$\text{dom } R \stackrel{\text{def}}{=} \left\{ x \in \bigcup \bigcup R \mid \text{there exists } y \text{ such as } x R y \right\} \quad (\text{domain of } R)$$

$$\text{ran } R \stackrel{\text{def}}{=} \left\{ y \in \bigcup \bigcup R \mid \text{there exists } x \text{ such as } x R y \right\} \quad (\text{range of } R)$$

Relations

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Note that $\text{dom } R$ and $\text{ran } R$ are sets defined from the Axiom Scheme of Comprehension:

$$\{ x \in A \mid \mathbf{P}(x) \}.$$

For using the convention on p. 11 the textbook write down these definitions by

$$\{ x \mid \mathbf{P}(x) \}.$$

Relations

Definition [2.]2.3

Let R be a binary relation. Then

$$\text{dom } R \stackrel{\text{def}}{=} \{ x \mid \text{there exists } y \text{ such as } x R y \} \quad (\text{domain of } R)$$

$$\text{ran } R \stackrel{\text{def}}{=} \{ y \mid \text{there exists } x \text{ such as } x R y \} \quad (\text{range of } R)$$

$$\text{field } R \stackrel{\text{def}}{=} \text{dom } R \cup \text{ran } R \quad (\text{field of } R)$$

Relations

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$$\text{field } R \stackrel{\text{def}}{=} \text{dom } R \cup \text{ran } R \quad (\text{field of } R)$$

If $\text{field } R \subseteq X$ then R is a relation in X .

Relations

Convention (p. 21)

When defining sets of ordered pairs we shall use the following convention:

$$\{ (x, y) \mid \mathbf{P}(x, y) \} \stackrel{\text{def}}{=} \{ w \mid w \doteq (x, y) \text{ for some } x \text{ and } y \text{ such that } \mathbf{P}(x, y) \}.$$

Relations

Definition [2.]2.11

Let A be a set. The **membership relation** on A , denoted by $\in_A$ is defined by

$$\in_A \stackrel{\text{def}}{=} \{ (a, b) \mid a \in A, b \in A \text{ and } a \in b \}.$$

Relations

Definition [2.]2.11

Let A be a set. The **identity relation** on A , denoted by Id_A , is defined by

$$\text{Id}_A \stackrel{\text{def}}{=} \{ (a, b) \mid a \in A, b \in A \text{ and } a \doteq b \}.$$

Relations

Definition

Let A and B be sets. The **Cartesian product** of A and B , denoted by $A \times B$ is defined by

$$A \times B \stackrel{\text{def}}{=} \{ (a, b) \mid a \in A \text{ and } b \in B \}.$$

Relations

Definition

Let A and B be sets. The **Cartesian product** of A and B , denoted by $A \times B$ is defined by

$$A \times B \stackrel{\text{def}}{=} \{ (a, b) \mid a \in A \text{ and } b \in B \}.$$

Remark

The above definition without using the conventions is

$$A \times B \stackrel{\text{def}}{=} \{ w \in \mathcal{P}(\mathcal{P}((A \cup B))) \mid w \doteq (a, b) \text{ for some } a \text{ and } b \text{ such that } \mathbf{P}(a, b) \},$$

where $\mathbf{P}(x, y)$: “ $x \in A$ and $y \in B$ ”.

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- Karel Hrbacek and Thomas Jech [1978] (1999). Introduction to Set Theory. Third Edition, Revised and Expanded. Marcel Dekker (cit. on p. 3).
- Casimir Kuratowski (1921). Sur la notion de l'ordre dans la Théorie des Ensembles. Fundamenta Mathematicae 2, pp. 161–171. DOI: [10.4064/fm-2-1-161-171](https://doi.org/10.4064/fm-2-1-161-171) (cit. on pp. 7, 8).